Ex - 09	**Model Deployment using REST API with Flask and Containerization with Docker**

AIM

To deploy a machine learning model using a Flask REST API and containerize the application using Docker.

PROCEDURE

1. Install Python, Flask, and the required machine learning libraries.
2. Create a simple machine learning model for prediction.
3. Save the trained model using the Pickle module.
4. Create a Flask application to load the saved model.
5. Define a REST API endpoint to accept input data and return predictions.
6. Test the API using a browser or a tool such as Postman.
7. Create a Dockerfile to package the application and its dependencies.
8. Build and run the Docker container to deploy the application.

PYTHON CODE

Save the following code as app.py.

```
from flask import Flask, request, jsonify
from sklearn.linear_model import LinearRegression
import numpy as np
import pickle

app = Flask(__name__)

# Train a simple model
```

```
X = np.array([[1], [2], [3], [4], [5]])
y = np.array([2, 4, 6, 8, 10])

model = LinearRegression()
model.fit(X, y)

# Save the model
pickle.dump(model, open("model.pkl", "wb"))
model = pickle.load(open("model.pkl", "rb"))

# REST API
@app.route('/predict', methods=['POST'])
def predict():
  data = request.get_json()
  prediction = model.predict(
    np.array(data['input']).reshape(-1, 1)
  )
  return jsonify({'prediction': prediction.tolist()})

if __name__ == '__main__':
  app.run(host='0.0.0.0', port=5000)
```

DOCKERFILE

Save this file as Dockerfile in the same folder as app.py.

```
FROM python:3.11-slim

WORKDIR /app
```

COPY app.py .

RUN pip install flask scikit-learn numpy

EXPOSE 5000

CMD ["python", "app.py"]

EXECUTION COMMANDS

Run these commands in the terminal from the project folder.

1. Install the required libraries

pip install flask scikit-learn numpy

2. Run the Flask application

python app.py

3. Build the Docker image (in another terminal)

docker build -t model-api .

4. Run the Docker container

docker run -p 5000:5000 model-api

5. Test the REST API

Send a POST request to http://127.0.0.1:5000/predict with the following JSON body using Postman:

```
{
  "input": [6]
}
```

OUTPUT

Flask server output:

* Running on http://127.0.0.1:5000

API response:

```
{
  "prediction": [12.0]
}
```

Docker output:

Successfully built image: model-api

Container started successfully.

RESULT

Thus, a machine learning model was successfully deployed using a Flask REST API and containerized using Docker for easy execution and portability.